

ST NAZAIRE MONTOR AP, France

WMO: 072170

Lat: 47.3139N

Lon: 2.1544W

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.5	-6.4	2.2	-1.3	-4.5	2.6	-0.4	13.7	11.4	12.4	11.0	2.4	30	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	30.0	19.8	27.7	19.1	25.7	18.3	21.1	27.2	20.1	25.5	19.3	23.8	4.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	23.0	18.3	13.2	21.9	17.6	12.6	21.1	61.1	27.4	57.4	25.4	54.8	23.8	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.8	8.7	-5.2	33.6	1.7	2.2	-6.4	35.2	-7.4	36.5	-8.4	37.7	-9.6	39.4
			-5.8	23.0	1.7	1.1	-7.0	23.8	-8.0	24.4	-9.0	25.0	-10.2	25.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	6.5	6.7	8.9	11.1	14.2	17.5	19.1	18.7	16.7	13.5	9.6	7.0	(6)
(7)		DBStd	5.51	3.52	3.26	2.86	2.77	2.81	3.03	2.70	2.63	2.73	3.01	3.28	3.76	(7)
(8)		HDD10.0	443	115	97	54	18	2	0	0	0	0	7	46	104	(8)
(9)		HDD18.3	2256	368	326	292	216	132	50	21	24	63	152	262	350	(9)
(10)		CDD10.0	1357	6	4	20	52	132	226	283	271	201	114	34	13	(10)
(11)		CDD18.3	127	0	0	0	0	4	26	46	37	14	1	0	0	(11)
(12)		CDH23.3	1061	0	0	0	3	42	236	380	282	113	4	0	0	(12)
(13)	CDH26.7	308	0	0	0	0	6	64	124	91	24	0	0	0	(13)	
(14)	Wind	WSAvg	4.1	4.4	4.5	4.5	4.3	4.0	3.9	4.0	3.7	3.7	3.9	4.1	4.5	(14)
(15)	Precipitation	PrecAvg	785	90	69	55	58	56	39	43	41	60	88	94	92	(15)
(16)		PrecMax	1008	178	153	161	159	117	80	93	108	172	186	189	160	(16)
(17)		PrecMin	514	10	4	6	15	13	4	4	2	2	18	27	6	(17)
(18)		PrecStd	128	49	37	37	35	25	23	21	28	41	42	45	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	15.1	20.0	23.5	27.3	32.0	33.0	32.9	29.7	23.4	17.8	14.7	(19)
(20)			MCWB	12.5	11.1	13.5	15.8	18.6	21.4	20.6	22.3	19.3	18.0	15.4	13.1	(20)
(21)		2%	DB	13.0	13.1	17.0	20.8	24.1	28.2	29.9	29.0	26.3	20.6	16.5	13.8	(21)
(22)			MCWB	12.0	10.9	12.1	14.7	17.4	19.5	19.6	19.8	18.1	16.9	14.7	12.6	(22)
(23)		5%	DB	12.2	12.1	14.8	18.4	21.7	25.9	27.5	26.2	24.1	19.1	15.5	13.2	(23)
(24)			MCWB	11.2	10.5	11.2	13.2	16.2	18.4	18.9	18.9	17.3	16.1	14.0	12.0	(24)
(25)		10%	DB	11.6	11.4	13.2	16.3	19.6	23.7	25.0	24.0	22.1	18.0	14.6	12.5	(25)
(26)			MCWB	10.5	10.0	10.6	12.3	15.2	17.5	18.2	18.1	16.7	15.7	13.1	11.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	12.2	14.1	17.1	19.9	22.4	22.0	23.2	20.7	18.7	16.2	13.7	(27)
(28)			MCDWB	13.4	13.5	18.5	21.7	25.6	29.3	29.2	30.6	26.2	21.5	16.9	14.2	(28)
(29)		2%	WB	12.2	11.6	12.8	15.3	18.0	20.5	20.7	21.1	19.4	17.7	15.2	12.9	(29)
(30)			MCDWB	12.8	12.6	15.5	19.6	22.8	26.2	27.0	26.9	23.9	19.6	16.1	13.6	(30)
(31)		5%	WB	11.4	11.0	12.1	13.8	16.7	19.3	19.8	19.9	18.5	16.9	14.3	12.3	(31)
(32)			MCDWB	12.1	11.9	13.9	17.4	20.6	24.5	25.4	24.3	22.1	18.5	15.4	13.1	(32)
(33)		10%	WB	10.6	10.3	11.3	12.7	15.7	18.1	18.9	18.9	17.7	16.2	13.2	11.5	(33)
(34)			MCDWB	11.4	11.3	12.9	15.7	19.0	22.6	23.5	22.4	20.8	17.7	14.5	12.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	6.8	7.8	8.8	8.9	9.7	9.6	9.8	10.3	8.0	6.9	6.1	(35)
(36)			MCDBR	5.2	7.1	10.7	12.3	12.5	13.6	14.6	14.0	13.4	8.9	6.6	5.2	(36)
(37)			MCWBR	4.6	5.5	6.9	7.2	6.8	6.3	6.1	6.3	6.4	5.8	5.3	4.8	(37)
(38)		5% WB	MCDBR	4.9	6.1	8.1	10.9	11.0	12.1	11.8	11.1	10.6	7.5	6.0	5.0	(38)
(39)			MCWBR	4.7	5.2	5.6	6.7	6.4	6.3	5.7	5.8	6.0	5.4	5.4	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.348	0.376	0.392	0.398	0.401	0.395	0.384	0.367	0.365	0.341	0.327	(40)	
(41)		taud	2.423	2.393	2.317	2.280	2.313	2.337	2.355	2.404	2.430	2.432	2.433	2.403	(41)	
(42)		Ebn at Noon	742	811	839	860	867	864	863	860	842	780	725	699	(42)	
(43)		Edh at Noon	69	87	109	124	124	122	118	108	96	83	68	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.09	2.01	3.21	4.76	5.63	6.20	5.93	5.22	4.14	2.38	1.37	0.95	(44)	
(45)		RadStd	0.10	0.20	0.34	0.48	0.39	0.54	0.28	0.38	0.28	0.19	0.12	0.11	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page